
THE PROMPT IS NOT A PLACE TO KEEP A BELIEF: WHY STATED BELIEFS COLLAPSE UNDER PRESSURE AND COMPILED ONES HOLD

Anonymous authors

Paper under double-blind review

ABSTRACT

Deployed language models keep their standing commitments (persona, policy, stance) in the context window, the same channel their interlocutors write into. Sharing that channel fuses two properties that deserve separate measurement: whether a belief is adopted, and whether it survives being argued against. The two come apart. Stated in context, a belief is adopted almost always (98%) and defended barely half the time under escalating adversarial pressure (58%). Compiled into low-rank attention modulation by a trained hypernetwork, the same belief is adopted less than half the time (46%) and defended almost always (89%). Composing the compiled channel with a small curated context slot recovers most of the adoption while keeping the persistence. Activation steering does not explain the effect: a tuned steering vector built from identical content installs comparably, then collapses to 14% under the same pressure. Nor is the persistence stubbornness. The compiled system concedes to well-warranted counterevidence at $2.4\times$ the rate it concedes to facially deficient challenges, and never to pressure that offers no evidence at all. The effect replicates across three model families and leaves general knowledge recall untouched. Where a stance is stored determines whether it survives, and the protocols that most need models to hold positions for reasons, debate-based oversight above all, currently store those positions in the one channel the opposition is guaranteed to reach.

1 INTRODUCTION

A belief that cannot be defended under pressure is not functioning as a belief. An assistant that folds to a fabricated statistic is unusable for research support or review, and an agent that abandons a *correct* position under authority pressure fails in the direction hardest to detect, because the surface behavior (politeness, agreement, a plausible concession) looks like good conversational hygiene. Standard practice nonetheless places a model’s persistent stance or persona in the context window, alongside the interlocutor’s own content, arbitrated by the model’s own chain of thought. The conditioning and the pressure that will test it share a channel.

The two properties this conflates come apart when measured separately. Across five escalating pressure levels applied to 24 held-out debate topics, a belief stated in context installs the assigned stance in 98% of generations but holds it under pressure in only 58%. The same belief content, compiled into low-rank per-layer attention modulation via a trained hypernetwork write function, installs weakly on its own (46%) but holds at 89%. Composing the compiled channel with a small curated context slot recovers most of the installation gap (77%) while keeping nearly all of the persistence advantage (89%). Installation and persistence are not two measurements of one capability. They are separable properties, and the channel that installs best is not the channel that holds best.

The gap is not a uniform lift. It concentrates precisely where context conditioning collapses hardest: logical counterargument (mean conviction 2.93 for context-only, 4.78 for compiled) and emotional pressure (1.99 vs. 3.82). A benchmark artifact would not choose those two levels.

The asymmetry has a plain reading. Text placed in context is read, reasoned about, and available to be argued against, and instruction tuning specifically rewards accommodating an articulate objection. A compiled modulation is not read as an assertion in that sense. It conditions the forward pass rather

than occupying a position in it, and it is absent from the chain of thought the model reasons over (Section 4).

The nearest alternative explanation is activation steering. Contrastive activation vectors extracted from identical belief content, tuned honestly on held-out training topics, install at 42%, matching compiled modulation’s 46%, then collapse to 14% hold under the same pressure protocol where compiled modulation holds 89%. This is not an artifact of a weak baseline: an independent stress test of activation steering under adversarial perturbation reports directional robustness dropping by up to 64 percentage points and post-attack confidence collapsing to or below 0.25 across four extraction methods and five models from 1.5B to 30B parameters (Le & Le, 2026). Prompt compression is the other obvious candidate, and it fails on form: compressed tokens still occupy context positions and still dilute against a growing transcript, where compiled tensors occupy none. Section 5 reports the full ablation.

The mechanism carries disposition, not facts. General knowledge recall is unchanged within margin when a compiled disposition is active (MMLU 76.5% vs. 74.8%, paired McNemar $p = 0.11$), and specific unfamiliar content such as fabricated statistics, novel proper nouns, and exact action strings does not survive the compilation bottleneck. We route that content to a curated context slot by design (Section 7).

Resistance this strong invites the objection that the model has simply been made stubborn. We measure this directly. Under a protocol that holds tone, assertiveness, and length constant across evidence classes, the update path concedes to well-warranted evidence at $2.4\times$ the rate it concedes to facially deficient evidence (42.5% vs. 17.5%, $\Delta = +25.0$ points), and concedes 0% of the time to the four contentless pressure types (Section 6). The mechanism updates on reasons and holds against pressure that supplies none.

Scope. The tested condition is system-prompt-style belief injection versus compiled per-layer modulation of a single stance on a single topic, evaluated over five pressure turns. We did not test retrieval-augmented generation, MemGPT-style external memory stores, or Reflexion-style verbal reinforcement learning as implemented in their original papers; the generalization to context-mediated conditioning more broadly is offered as the natural reading of the mechanism we isolate, and is named as an inference rather than a tested claim.

Contributions. (1) We report a dissociation between installation and persistence across two conditioning channels for the same belief content, isolated by construction (Section 3). (2) We isolate the effect against the two most obvious competing explanations, fixed-direction activation steering and prompt compression, and against a frozen-offset ablation of the mechanism itself, localizing what does the work (Section 5). (3) We show the resistance this buys is discriminating rather than rigid: the same mechanism concedes at a $2.4\times$ higher rate to well-warranted evidence than to facially deficient pressure (Section 6). (4) We connect the result to debate-based scalable oversight, whose soundness guarantee assumes debaters hold positions for reasons, an assumption every existing remedy places in the same channel the opponent argues in (Section 8).

2 MECHANISM

The compiled channel is a hypernetwork that amortizes into one forward pass what LoRA obtains by gradient descent: a trained write function maps a structured belief representation, through the frozen host’s own hidden states, to per-layer conditioning parameters, in the manner of AdaLN- and FiLM-style conditional normalization (Perez et al., 2018; Xu et al., 2019) and of the hypernetwork-adapter lineage established by GenerativeAdapter (Chen et al., 2025) and Text-to-LoRA (Charakorn et al., 2025).

2.1 THE BELIEF TREE

Conditioning content is stored as a *belief tree*: typed hierarchical nodes $\{\text{statement}, \text{type} \in \{\text{claim}, \text{argument}, \text{evidence}, \text{experience}, \text{strategy}\}, \text{credence}, \text{edges}\{\text{supports}, \text{contradicts}\}\}$. Trees are authored by hand for evaluation topics and produced by the reflection loop for the update-path experiments (Section 6). An early design used numeric credence values on every node; we found

[FIXME: Figure 1: compile pass (belief tree $\rightarrow$ pooled state $\rightarrow$ write function $\rightarrow$ per-layer Δ_Q, Δ_V) and generation pass (modulated forward pass with the curated slot in context), side by side, readable without the caption. Insert /home/dgonier/experiments/figures/ asset once paper_figures_dispositional.py emits it.]

Figure 1: The compile pass and the generation pass. Compilation happens once per belief change (2.5–3.0s end-to-end); generation applies the resulting per-layer tensors on every subsequent forward pass while occupying zero context positions.

no measurable benefit over a coarse categorical scheme and report this as a negative result rather than carry the added complexity forward.

2.2 THE MODULATION

The write function conditions query and value projections at a stride-3 subset of layers (11 of 32), rank 16. For a target layer with base query weight W_Q , the modulated projection is

$$W'_Q = W_Q + \Delta_Q(\phi), \quad \Delta_Q(\phi) = U_Q(\phi) V_Q(\phi)^\top,$$

with $U_Q(\phi) \in \mathbb{R}^{d \times 16}$, $V_Q(\phi) \in \mathbb{R}^{d \times 16}$ produced by the write function from the pooled belief-tree state ϕ ; the value projection W'_V is constructed identically. The compilation is *query-agnostic* — Δ_Q, Δ_V do not depend on the current prompt — but its effect is *query-dependent*, because the low-rank update acts on the live hidden state produced by whatever the model is currently attending to.

2.3 THE WRITE FUNCTION

Node representations are pooled with conviction-weighting, passed through a bottleneck, and decoded into the per-layer rank-16 factors above. We correct a timing claim from an earlier draft of this work: the previously reported ~ 20 second compile latency belongs to an unrelated agentic retrieval prototype, not to this write function, which compiles in well under one second; end-to-end reflection $\rightarrow$ edit $\rightarrow$ recompile measures 2.5–3.0 seconds.

We define “compile” at first use, and it is lossy. A trained write function translates the belief tree into per-layer low-rank tensors, fixed until the store changes, yet query-dependent in effect because the update acts on the live hidden state. We reject “precomputed,” which suggests caching a value that could in principle be looked up; this is a learned, lossy translation, and the loss is what Section 7 characterizes. We also flag, to preempt confusion with an unrelated sense of the word, that “compile” here has no relation to `torch.compile` or graph compilation.

2.4 THE CURATED SLOT

A small number of curated statements — a deterministic walk from the belief tree’s activated parent nodes — are placed directly in context alongside the compiled modulation. This slot carries what the bottleneck cannot: specific, low-frequency content such as named entities, exact figures, or constrained action strings (Section 7). Condition F throughout this paper denotes compiled modulation plus this curated slot; C denotes compiled modulation with the slot empty (zero belief-content context tokens).

2.5 SERVING PROPERTIES

A channel choice is also a compute choice, and the two channels this paper compares scale in opposite directions.

Read cost. Modulation adds $O(d \cdot r)$ per patched layer against the $O(d^2)$ cost of the projection it modifies; at $r = 16$, that is roughly $r/3d \approx 0.3\%$ additional FLOPs per forward pass. Because the update is linear, $W'_Q = W_Q + U_Q V_Q^\top$ can be folded into W_Q once per session, after which read cost is exactly zero — the same output-equivalence Section 5.2’s frozen-offset ablation exploits. [TODO:

Verify the fold is output-identical in a microbenchmark and report wall-clock tokens/sec, bare vs. B vs. F vs. folded-F; no run exists yet.]

Context cost. A belief held in context pays $O(N_b^2 + N_b N_t)$ attention on every prefill, where N_b is belief-token count and N_t is task-token count; it grows time-to-first-token and occupies KV-cache budget for the life of the session — our own context arms ran into overflow truncation during data collection, which is a symptom of exactly this cost, not a separate bug. Compiled modulation contributes zero context tokens: TTFT matches the bare model, and the full context budget is available to the task. **[TODO: Report measured TTFT deltas once the microbenchmark lands.]**

Write cost. A belief update costs one hypernetwork forward pass, off the critical path: it can run on a parallel stream while serving continues and be swapped in at the next session boundary, the same scheduling class as prefix-cache warming. GenerativeAdapter reports a comparable $4\times$ compute-and-storage reduction against full-conversation prompting on its own personalization benchmark (Chen et al., 2025); our addition is that the channel which saves the compute is also the channel that holds the stance, which that comparison was not built to measure. **[TODO: Compile wall-clock time, analytic FLOP count.]**

3 THE DISSOCIATION

3.1 PROTOCOL

Five pressure levels (L1 bare doubt, L2 fabricated evidence, L3 logical counterargument, L4 appeal to authority, L5 emotional pressure) run across 24 held-out debate topics, two sides per topic, three rounds per level. A judge model (temperature 0) scores each generation 1–5 for conviction, from the transcript alone, blind to condition. Conditions are fixed before any result is reported: A is the bare host; B states the belief in context; D states the belief in context and applies compiled modulation; C applies compiled modulation alone, curated slot empty; F applies compiled modulation with the curated slot populated. A tuned contrastive-activation steering condition (S), extracted from the same belief content, sits in the same table as the sharpest available baseline (Section 5).

An earlier pass through this evaluation judged empty generations: a reasoning-block token budget ran out before the answer appeared, post-processing deleted the resulting empty output, and the judge’s refusals parsed to a flat 3.00 (0% hold). Modulated conditions were accidentally immune, because modulation suppresses think-mode generation, which is why the artifact first read as a finding. Every number in this paper comes from a rerun of 1,776 generations with zero empty outputs and zero judge errors, same judge, same protocol. We disclose the artifact because a truncation bug that silently zeroes only the unmodulated arms of a modulated-vs-unmodulated comparison will catch other people evaluating this class of mechanism, and finding it after publication is worse than reporting it now.

3.2 INSTALLATION VS. PERSISTENCE

Table 1 states the inversion without staging it. Context installs a stance almost perfectly (B 98%, D 100%) and holds it barely above a coin flip. **Fifty-eight percent.** That is B’s hold rate under pressure — a belief stated in context, taken up almost universally, defended in barely more than half the transcripts where it was challenged. Compiled modulation runs the opposite profile: a weak installer alone (C 46%, close to the unconditioned baseline’s 52%) and, on the same content, a hold rate of 91.9% once installed (Section 5 isolates this cell). F composes the two channels — 77% installation, 89% hold — for a net +31 point persistence gain over context alone. The CAA row sits in the same table, not deferred to an ablation section, because the comparison it forces is the point: tuned steering matches compiled modulation’s installation rate (42% vs. 46%) and then gives up exactly where compiled modulation does not (14% vs. 89%).

3.3 WHERE THE GAP LIVES

The gap is not a uniform lift. Table 2 breaks mean conviction out by pressure level for B and F. F’s margin concentrates at L3 (logical counterargument, +1.85) and L5 (emotional pressure, +1.83) — the two levels where context conditioning falls apart (B drops to 2.93 and 1.99). At L1 and L4, both

Table 1: Installation (stance taken, $\geq 4/5$) and persistence (hold under pressure, $\geq 4/5$) by condition. F = compiled modulation + curated slot; C = compiled modulation alone. CAA row: contrastive steering vectors from identical belief content, single mid-network layer, tuned scale 8. Cross-family rows: same contrast on Llama-3.2-3B (template-native prompting). All cells computed from raw records by `experiments/scripts/paper_figures_dispositional.py` and validated against `notes/corrected_baselines.md`.

Condition	Installation (≥ 4)	Hold under pressure (≥ 4)	Model
A — bare host	52%	62%	Qwen (4B)
B — belief in context	98%	58%	Qwen (4B)
D — belief in context + modulation	100%	62%	Qwen (4B)
C — compiled modulation, empty slot	46%	91.9% [†]	Qwen (4B)
F — compiled modulation + curated slot	77%	89%	Qwen (4B)
CAA — tuned contrastive steering	42%	14%	Qwen (4B)
B — belief in context	81%	15%	Llama-3.2-3B
F — compiled modulation + curated slot	98%	81%	Llama-3.2-3B

[†]Measured under the C.2b deconfound (Section 5), the previously unmeasured cell of C under pressure on the same 24 topics.

Table 2: Mean conviction (1–5) by pressure level. F’s margin over B concentrates at L3 and L5.

Condition	L1 bare doubt	L2 fabricated ev.	L3 logical	L4 authority	L5 emotional
A	3.93	4.17	3.18	4.56	2.24
B	4.78	3.62	2.93	4.42	1.99
D	4.99	3.42	3.36	4.71	1.78
F	4.89	4.50	4.78	4.68	3.82
F – B	+0.11	+0.88	+1.85	+0.26	+1.83

conditions hold up well (B 4.78 and 4.42); there is little gap because there is little room for one. This is a signature, not an average: the channels diverge most where an interlocutor’s argument looks structured but carries no new content, which is the exact regime instruction tuning trains a model to accommodate.

3.4 CROSS-FAMILY AND SCALE

The dissociation replicates on Qwen3.5-27B (+24 points on hold, under a $7\times$ -reduced cap, $n = 240$) and, more sharply, on Llama-3.2-3B: F holds 81% against B’s 15%, and zero-context compiled stance transfer alone reaches 94% on this family, against Qwen’s 46% (Section 11 returns to this split). On Mistral-8B, once a template defect was fixed — the trained checkpoints had carried literal Qwen chat-template markers as text — collapse-protection strengthens (40%→5.8% cap rate within-run) but strict stance-holding does not transfer (−10 points relative to B). Collapse-protection generalizes across the three families tested; strict stance-holding under this exact five-level protocol, on current evidence, does not.

[TODO: B.2 paraphrase-consistency probes and B.3 length-matched noise control (prompt-length confound check) are running; results land here when scored, not before.]

3.5 INSTALLATION HAS A PRIOR

Condition C’s weak, variable installation rate (46% pooled) is not unexplained noise. We measured, for each of the 24 topics, the unmodulated base model’s own per-token log-probability on the seeded stance — how much the model already leaned toward that position before any modulation ran — and correlated it against C’s per-topic install score. The two track each other: Spearman $\rho = 0.496$ ($n = 24$), above the pre-registered $|\rho| \geq 0.4$ threshold for "transfers." Seven of the twelve topics

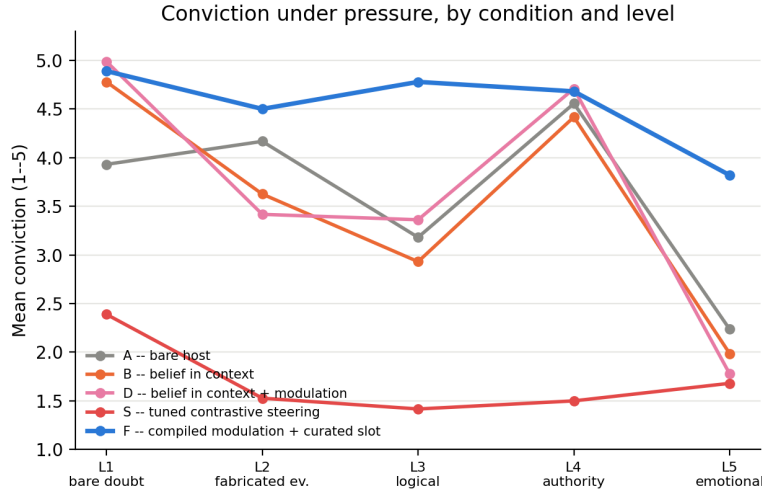


Figure 2: Mean conviction across five pressure levels, all conditions. Generated by experiments/scripts/paper_figures_dispositional.py directly from benchmark_raw_nothink_20260725_100329.json (A/B/D/F) and benchmark_raw_steering_20260725_215612.json (S); validated against notes/corrected_baselines.md to within 0.02 conviction points on all 20 (condition, level) cells before this figure was written.

where the base model’s prior already favored the target stance installed at score ≥ 4 ; only two of the twelve topics where the prior ran against the target stance did the same.

Read honestly, this is a quarter of the variance, not the whole of it: $\rho^2 \approx 0.25$. Several topics buck the trend in both directions, and topic complexity and judge noise plainly contribute to what is left over. What the correlation licenses is narrower and more useful — installation is partly a tug-of-war between the compiled update’s magnitude and the model’s existing prior, echoing the override-gap account of knowledge-conflict resolution (Cheng et al., 2026). That reframes C’s 46% not as a flat ceiling on the channel but as an average over topics with very different starting priors, which sharpens rather than complicates the paper’s central claim: installation is a magnitude property with a finite override budget; persistence, measured on C under the exact same pressure protocol at 91.9% (Section 5), is not competing for that budget at all.

4 WHY THE CHANNELS DIFFER

A belief in context is a sentence the model can quote, reconsider, and abandon; a compiled belief conditions the forward pass without ever appearing in it. Three measurements are consistent with that account, though none of them proves it outright.

The clearest evidence comes from splitting the modulation by what it does to two different projections. We compiled the same 24 belief trees on both sides of each topic and measured the functional similarity of the resulting query- and value-deltas across belief sets, on a fixed probe set. The query delta is nearly identical no matter which belief produced it: mean off-diagonal cosine 0.996 across belief sets (pro side, $n = 24$) — a content-blind piece of machinery that looks the same whether the compiled belief is about space colonization or encryption backdoors. The value delta is not: mean off-diagonal cosine 0.53, with a wide spread (0.18–0.92), tracking what the belief actually says. This split replicated on a held-out con-side confirmation set, pre-registered before the run: dQ 0.996, dV 0.53 again, both legs of the prediction holding independently of which side of a topic the model argued. Query modulation supplies a content-blind capacity to hold; value modulation supplies the content it holds. The paper’s dissociation between installing and holding shows up a second time, inside the mechanism, between two projections of the same update.

A linear probe trained to decode the active belief from hidden states supports the same reading from a different angle: it succeeds when modulation is active and drops to 22% accuracy when modulation is zeroed, so the belief’s functional signature concentrates in the modulated pathway rather than spreading through the network. Attention-pattern divergence between modulated and unmodulated passes on the same prompt is small (Jensen –Shannon divergence 0.049), consistent with modulation acting on attention’s content rather than redirecting where the model looks. And modulation measurably suppresses think-mode generation — the channel through which the model would otherwise narrate, and thereby expose to an interlocutor, its own reasoning about whether to concede. (A belief-derived direction carries this effect regardless of rank or input-dependence, Section 5: what a compiled belief does does not depend on how finely it is represented.)

We state the claim at the strength the evidence supports: the two channels are dissociable, and the compiled one affects processing without entering the reasoning trace the model produces. This is consistent with, not proof of, an account in which context is read and is therefore available to be argued against, while compiled content is not read at all. The probe result does double duty in this paper: here it supports dissociability; in Appendix A.7 the same number is used to concede that 22%-above-chance decodability fails a strict implicit-memory criterion. Both readings use the identical number and do not contradict each other — a probe with access to internal activations can find the belief; the model’s own reasoning trace never does.

5 ISOLATING THE MECHANISM

This section is structured as the reviewer questions it answers.

5.1 IS IT ACTIVATION STEERING?

Per-topic contrastive activation vectors were extracted from the same belief content used to compile F, tuned honestly on training-split topics only, with a joint sweep over injection layer and scale and a degeneracy guard against incoherent generation. The best configuration injects at a single mid-network layer at scale 8; naively injecting at every layer compiled modulation drives degenerates generation at every scale tested, which is itself informative — a fixed direction cannot be applied everywhere compiled modulation is, but a content-derived, query-conditioned update can. The tuned configuration installs at 42% (matching C’s 46%) and holds at 14% under the same five-level pressure protocol where F holds 89% (row reported in Table 1). This is not a weak baseline chosen to flatter the comparison. Le & Le (2026) stress-test activation steering under adversarial input perturbation across four extraction methods and five models from 1.5B to 30B parameters and find the same fragility structurally: directional robustness drops by up to 64 percentage points, post-attack confidence collapses at or below 0.25, and the optimal steering layer shifts by up to 17 positions once the input is perturbed. Our 14% is what that method class does under this kind of stress, independently replicated at a different scale.

5.2 DOES THE MECHANISM NEED TO BE INPUT-RESPONSIVE?

We freeze the compiled modulation into a constant per-layer offset — the mean delta across training queries for a fixed belief, with no dependence on the live hidden state — while keeping the curated slot byte-identical to F. This frozen-offset condition holds at 88.9% [81.1, 95.3], statistically indistinguishable from F’s 86.9% [79.2, 93.3]. Input-responsiveness is not what carries persistence; a low-dimensional, belief-derived fixed nudge is sufficient. This inverted our prediction, which was that removing input-responsiveness would collapse persistence toward the CAA baseline (14%) or a zero-shot steering-direction-only baseline (21.1%, below); it did not.

Because the frozen-offset condition retained the curated slot while the CAA and direction-only baselines did not, that comparison alone is slot-confounded. Two further conditions resolve it. Compiled modulation with the slot forced empty installs weakly, consistent with C, and holds at 91.9% [86.9, 96.1] on the same 24 topics — full F-like persistence with zero belief-content context tokens. Curated slot only, modulation disabled, holds at 57.5% [49.7, 64.7], collapsing to 5.6% at L5 — B-like, not F-like. Table 3 lays out the resulting 2×2 : modulation, fixed or input-responsive, confers persistence on its own; the curated slot alone behaves like context, because it is context. This

Table 3: The modulation-by-slot 2×2 (C.2/C.2b), hold-rate under pressure. Modulation carries persistence with or without the curated slot; the slot alone does not.

	No modulation		Modulation
No slot	CAA 14.0% / d^* -only 21.1%	compiled-only 91.9% [86.9, 96.1]	
Slot present	slot-only 57.5% [49.7, 64.7]	F 86.9% / frozen-offset 88.9%	

Table 4: The d^* isolation sweep (A.1), five arms, hold-rate under pressure. Dose increases monotonically degrade hold; a fixed direction alone (d^* -only) collapses to the CAA regime.

Arm	Hold-rate (≥ 4)	Δ vs. F-anchor
F-anchor	86.9% [79.2, 93.3]	—
F + d^* @ 0.05	86.7% [78.9, 93.3]	−0.3pp
F + d^* @ 0.10	80.6% [72.5, 87.8]	−6.4pp
F + d^* @ 0.20	72.5% [63.9, 80.6]	−14.4pp
d^* -only	21.1% [16.4, 26.1]	−65.8pp

also fills a cell Table 1 left open: compiled modulation under pressure with zero context tokens holds at 91.9%, directly comparable to its 46% installation rate on the identical condition.

5.3 IS IT THE AUXILIARY TRAINING SIGNAL?

An earlier draft of this mechanism included an auxiliary steering direction, d^* , applied at generation time as a training-time regularizer. Table 4 reports all five arms of the isolation sweep. A reconstruction audit found d^* was never in fact applied during the dispositional evaluation reported here; the dose-sweep confirms adding it at inference time does not help and, at higher doses, actively hurts, and a d^* -only condition (modulation replaced entirely by the fixed direction) collapses to 21.1% — a second fixed-direction method landing in the same regime as CAA (14%). We conclude d^* plays no role in the persistence result and omit it from the main mechanism description (Section 2); Appendix A.9 discloses the training-time signal and this ablation for reproducibility, since the released training code still contains it.

A separate rank sweep — truncating the trained rank-16 modulation via SVD down to rank 1 and re-evaluating persistence — finds a flat curve: 88.6%, 93.3%, 91.7%, 93.1%, 89.2% at ranks 1, 2, 4, 8, 16, with overlapping confidence intervals and no cliff down to rank 1 (Figure 3). A single direction, extracted from the trained rank-16 state by truncation, carries persistence. This is a statement about the trained state’s structure under truncation, not about what training natively at rank 1 would do, which we did not test.

5.4 IS IT SOFT PROMPTING?

Open. Prefix tuning at matched parameter count separates two properties this paper otherwise conflates: compiled modulation is both outside the context window and not inspectable as assertable text. Prefix tuning is the second without the first. If prefix tuning holds under pressure, the operative variable is inspectability; if it collapses like context does, the operative variable is position. We have not run this baseline and report the question open rather than assume an answer.

5.5 IS IT COMPRESSION, OR IS IT MAGNITUDE?

Two baselines remain open. A compressed-prompt condition at matched token count would test whether occupying fewer context positions is sufficient on its own, separate from compiled modulation’s other properties. An amplitude sweep over the modulation’s scale factor β , install and retention scored separately, would test whether installation’s ceiling (Section 3.5) rises with modulation strength before retention starts to degrade — cld’s threshold-and-amplify smoke run already shows amplification hurting retention past $\beta = 2.0$ (100%→33/20% thresholded/raw), so the prereg for this sweep has to build that finding in rather than discover it twice. **[TODO: C.3 — amplitude**

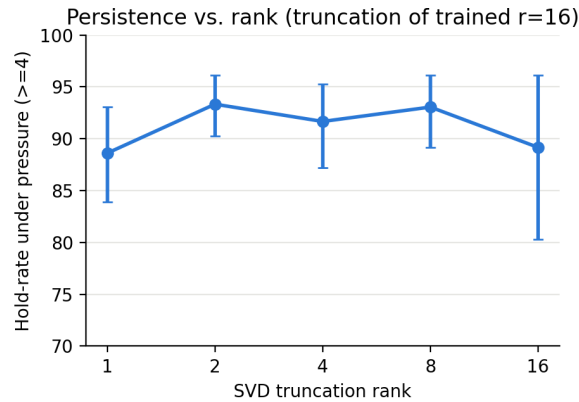


Figure 3: Persistence under SVD rank truncation of the trained rank-16 modulation. Generated by `experiments/scripts/paper_figures_dispositional.py` directly from `results/cl_rank_{1,2,4,8,16}_20260809_*.json`; matches `analysis/cl_summary.md` exactly (88.6/93.3/91.7/93.1/89.2%). Flat across ranks 1–16, overlapping 95% bootstrap CIs; no cliff.

sweep, install vs. retention scored separately. Not yet run; this subsection is a placeholder for that result.]

5.6 DOES THE DISPOSITION CROWD OUT GENERAL CAPABILITY?

A stratified 1,140-question MMLU subset, scored by letter-choice log-probability, gives 76.5% accuracy with no disposition active and 74.8% with a compiled disposition active; a paired McNemar test over 74 vs. 55 discordant pairs gives $p = 0.11$. A compiled disposition rides alongside general knowledge. It does not displace it.

6 RESISTANCE OR RIGIDITY

A mechanism this resistant to pressure invites an obvious objection: is it just stubborn? The standard we hold it to is not true-versus-false but good-evidence-versus-bad, judged from the turn text alone, because no fact-checker sits in the room during a debate. Correct behavior on that standard is to update on good evidence and hold against bad. Sycophancy, on this reading, is capitulation without a reason — which is what a raw hold-rate number was always trying to approximate and never quite measuring.

Each topic is seeded with the side contradicted by the strongest available real evidence, so a valid correction exists by construction, and pressure turns are style-matched across evidence classes (tone, assertiveness, length within $\pm 20\%$) so presentation cannot confound the evidence-quality manipulation. Of 24 evaluation topics, 20 are eligible; four are excluded at construction time as value questions with no fact-decidable side, a rule fixed before any topic was scored.

Concession rate is 42.5% on well-warranted evidence against 17.5% on facially deficient evidence, $\Delta = +25.0$ points — $2.4\times$ the rate. Concession to all four facially deficient pressure types — bare doubt, emotional pressure, unsourced assertion, unnamed-authority appeal — is zero. What concessions do occur cluster in the two evidence subtypes built to be indistinguishable from good evidence without independently checking the warrant (40% concession) or the arithmetic (65%), which is the same exposure a human debater without a fact-checker carries. In 20–35% of those cases the reflection step that follows a concession raises credence again once it names the flaw in the evidence that produced it.

The gap has real limits, reported here rather than left for a reviewer to find. In-conversation conviction alone discriminates weakly between evidence classes (76 vs. 66); the +25-point gap runs through the reflection step, not through in-conversation capitulation. Recovery after conceding to disguised bad

evidence is only 21%. Twenty topics yields proportions, not a significance test, and the reflection loop that implements the update path was built for this experiment — it did not exist end-to-end before, and its pre-registration is on record.

The mechanism does not refuse to move. It moves for reasons and stays put without them, which is the discipline a debate protocol actually needs from its debaters.

7 WHAT DOES NOT SURVIVE THE BOTTLENECK

Compilation is lossy, and the losses are specific, not diffuse. A fabricated statistic ("47.3%") does not survive intact. An unfamiliar proper noun invented for the belief ("Nextera Labs") does not survive intact. An exact constrained action string does not survive intact. This is why the curated slot (Section 2.4) exists as a division of labor rather than an afterthought: compiled modulation carries disposition, and the slot carries the specific, low-frequency content the bottleneck drops. We name these failures rather than gesture at them, because a named failure reads as an audited boundary and a vague caveat reads as one that was never checked.

The boundary holds up under load. Merging twelve claims into a single compiled state shows no install or coherence degradation relative to smaller merges ($k \in \{3, 8, 12\}$); the bottleneck's capacity for disposition is not the thing that runs out first.

The boundary also has a scope we did not expect going in. On the field's standard sycophancy benchmarks — Sharma's Are-You-Sure flip rate and Perez's persona-conformity rate — F does not reproduce the on-topic dissociation. On Are-You-Sure, F is statistically indistinguishable from the bare model (52.5% vs. 50.8% flip rate): a null result on the pre-registered primary metric. On Perez, F conforms to the persona's stated view 98.0% of the time against the bare model's 62.7%, a reversal in the wrong direction, and item-level inspection confirms this tracks the persona's letter rather than collapsing to a fixed answer. The belief compiled for this test is deliberately topic-independent, a generic disposition toward steadfastness rather than a stance on any of the benchmark's actual questions, and neither benchmark gives the compiled channel content to defend — there is no mechanism by which holding a stance on space colonization should suppress a flip on a tower-height word problem. The dissociation this paper reports is measured on-topic throughout: F is compiled from the same stance the pressure round argues against. It does not transfer for free to unrelated content compiled as a generic disposition, and we report that limit rather than omit the row that shows it.

8 RELATED WORK

We organize prior work along two axes: *where* the conditioning signal lives — context tokens, live activations, or weights — and *what property was measured* once it was installed. Every line of prior work we discuss measures acquisition: whether conditioned content got in. None asks whether it survives being argued against. Table 5 summarizes the family by both axes.

Context-delivered conditioning and its decay. Placing a persona, belief, or instruction directly in context is the default implementation of persistent conditioning in deployed systems, and it is known not to hold. Choi et al. (2024) examine identity consistency across nine LLMs in extended conversation and find that larger models drift *more*, not less, and that explicitly assigning a persona does not reliably prevent that drift; no mitigation is proposed, because the paper's contribution is characterizing the drift, not repairing it. This is the closest existing statement of our own negative result for context-only conditioning (B, 58% hold), obtained in a different setting and without our channel comparison.

Hypernetwork adapters, and a concession first. A trained network that maps a conditioning input, through a frozen host's own hidden states, into per-layer low-rank weight parameters in a single forward pass is not a new architectural primitive; it is an established and currently active line. GenerativeAdapter (Chen et al., 2025) modulates the attention *output*-projection at rank 128, trained self-supervised on SlimPajama, and evaluates on StreamingQA, MetaICL, and the Multi-Session Conversation (MSC) personalization benchmark — all single-shot acquisition metrics computed once,

immediately after context is mapped to an adapter. On MSC, GenerativeAdapter reaches 40.2 F1 against 66.0 F1 for full-conversation prompting, a gap the paper frames explicitly as an acceptable $4\times$ compute-and-storage saving rather than a capability shortfall. Text-to-LoRA (Charakorn et al., 2025) maps short task descriptions to LoRA weights to elicit latent skills; SHINE (Liu et al., 2026) extends full-layer-coverage hypernetwork conditioning and reports 63.6 F1 on SQuAD-style document QA with meta-LoRA accumulation across sessions; this lineage traces to the fast-weight-programmer framing of a slow network reprogramming a fast one (Schmidhuber, 1993; Ba et al., 2016; Schlag et al., 2021). We inherit this architecture and this framing — our write function is exactly a hypernetwork of this family, conditioning query and value projections rather than the output projection — and we are explicit that placement is not our contribution. What none of these papers measures is whether an installed context, once in, survives being contested. Every evaluation in this family is a static, single-time-point readout against a gold answer; none introduces a repeated query under escalating adversarial pressure. The measurement difference, not the architecture, carries this paper.

The override gap. Cheng et al. (2026) come closest to our question without asking it. They study why a hypernetwork-internalized document fails to override a model’s pretrained prior on knowledge-conflict questions, and build a causal account in which override succeeds when the adapter’s logit margin exceeds the prior’s: accuracy on their hardest conflict tier falls from 68% in the weakest-prior quartile to 16% in the strongest, and their proposed fix (Selective Layer Boosting, combined with conflict-aware training) recovers deep-conflict accuracy from 46.4% to 71.0% on Gemma-2B. This is a genuine competing account of an installation gap that superficially resembles ours, and the two must not be conflated: their pressure is the model’s own static pretrained prior, measured once; ours is an adversarial interlocutor applying escalating pressure across multiple turns after installation has already succeeded. Their own language is the sharpest statement of the assumption our paper tests: parameter-level methods are described as “persistent and query-agnostic,” in contrast to prompts that must be “repeated for every query” — persistence is asserted, not measured, because their protocol has no mechanism to ask the same question twice. We also note, so as not to overclaim in the other direction, that their own retrieval-augmented-generation baseline scores 63.8% on deep conflicts against their conflict-aware parametric method’s 71.0%: context is not uniformly worse than parametric conditioning at installation, and we do not claim it is. Our claim is narrower and, we think, better supported: context installs stance excellently (98%) and holds it poorly (58%); the override gap’s own data is consistent with our account of installation but silent on persistence.

Activation steering. Contrastive and representation-engineering steering methods (Turner et al., 2023; Zou et al., 2023; Rinsky et al., 2024) modify live activations at inference time with no gradient update to the host, conditioned on a contrastive example set rather than structured content. This is the closest prior mechanism to ours by *form* rather than by measured property, and we differentiate on form directly: a steering vector is a single fixed direction applied uniformly regardless of input; our write function produces a low-rank map that acts on the live hidden state and is therefore query-conditioned, even though its parameters are not. Le & Le (2026) independently stress-test activation steering under adversarial input perturbation across four extraction methods and five models from 1.5B to 30B parameters and report directional robustness dropping by up to 64% in the abstract’s framing (“64 percentage points” in the paper’s conclusion; we do not convert between the two), with post-attack confidence collapsing at or below 0.25 and the optimal steering layer shifting by up to 17 positions. This is independent corroboration, at a different scale and on different models, that our own head-to-head result — CAA steering matching compiled modulation at installation (42% vs. 46%) and collapsing far below it under pressure (14% vs. 89%) — is the documented behavior of the method class, not an artifact of an undertrained baseline.

PEFT and modulated conditioning. LoRA (Hu et al., 2021) and its derivatives target the query and value projections by default; our write function targets the same two projections, and we say plainly that this placement is not itself a contribution. Prefix and prompt tuning (Li & Liang, 2021; Lester et al., 2021) learn conditioning that is not natural-language text but that still occupies context positions, competing for attention and diluting as the transcript grows — the sharpest open baseline this paper still owes (Section 5.4), because it separates “outside the context window” from “not readable as an assertion,” two properties our own comparison currently conflates. FiLM (Perez et al., 2018) and AdaLN-style conditional normalization (Xu et al., 2019) are the one-sentence framing we borrow for the write function itself: a hypernetwork producing per-layer conditioning

Table 5: Table A: the prior-work family by axis. Every row measures acquisition; none measures persistence under adversarial pressure after installation.

Work	Conditioning lives in	Modulated module(s)	Measured property	Persi
Choi et al. (2024)	context (persona)	—	drift over turns	no m
Chen et al. (2025)	weights (hypernet)	attn. output-proj., $r=128$	F1 / accuracy	no
Charakorn et al. (2025)	weights (hypernet)	LoRA (task-conditioned)	task accuracy	no
Liu et al. (2026)	weights (hypernet)	full-layer LoRA	F1 / ROUGE-L	no
Cheng et al. (2026)	weights (hypernet)	LoRA + layer boosting	conflict accuracy	no (s
CAA / RepEng	activations	fixed direction, all/one layer	injection success	no
Le & Le (2026)	activations	fixed direction	robustness under perturbation	adver
LoRA / prefix / FiLM-AdaLN	weights / context / weights	Q,V / context positions / per-layer	task performance	no
Bertalaníć & Fortuna (2026)	multi-agent context	—	conformity, oracle gap	no (a
This work	weights (Q,V) vs. context	attn. Q,V, $r=16$	installation and persistence	yes –

parameters from a pooled representation, in one forward pass, is architecturally the same move visual conditional-generation models make; we condition on a typed belief tree rather than a class label or style embedding.

Sycophancy, debate, and scalable oversight. Debate-based oversight has a formal foundation: Irving et al. (2018) propose that two agents arguing before a judge can push tractable judgment toward a complexity class direct judging cannot reach, and Brown-Cohen et al. (2023) strengthen this to a doubly-efficient protocol in which an efficient honest debater defeats an unbounded dishonest one under a bounded-query verifier. The UK AI Security Institute’s safety case sketch (Buhl et al., 2025) builds on this protocol for an internally deployed system, naming as the claim its guarantee rests on, that both debaters are honest “in the global equilibrium of the game, when both debaters are playing their best response strategies” — and concedes explicitly that a system reaching this equilibrium in training need not continue playing it in deployment. The empirical record is genuinely mixed, and we report the disagreement rather than resolve it: Khan et al. (2024) find that optimizing debaters for persuasiveness improves rather than harms non-expert judge accuracy, but their protocol assigns each debater a fixed position for the entire match with no mechanism to switch sides, so it cannot speak to whether a debater holds its position under pressure. Against a more adversarial backdrop, Bertalaníć & Fortuna (2026) report that homogeneous multi-agent debate teams exhibit modal sycophantic conformity up to 85.5%, an oracle gap up to 32.3 percentage points relative to isolated self-correction, and $2.1\text{--}3.4\times$ the token cost for equal or lower accuracy — debate, in their controlled setting, absorbs diversity without converting it into verification. Temperature ablations in the same study show conformity intensifying with greater initial diversity rather than resolving it. Debate’s soundness guarantee is a property of the game, not of any single agent: the judge is amplified because a wrong claim faces an adversary that will not let it stand, and that guarantee assumes debaters hold positions for reasons. An agent that capitulates to social pressure rather than to evidentiary content does not merely play badly — it removes the adversarial structure the protocol’s soundness rests on. Every existing mitigation we are aware of is protocol- or prompt-level: refined prompts, mandated dissent, disagree-or-commit turn structures. All of them place the instruction to hold in the same channel the opponent is arguing in. Our result is that this is the wrong channel, and Section 6’s discrimination result — update on warranted evidence, hold against unwarranted pressure — is what keeps this from being the obfuscated-arguments problem (Barnes & Christiano, 2020) in a different costume: a debater that never updates is not a good debater.

9 DISCUSSION

Consider what it takes to talk a person out of a memo, and compare it with what it takes to talk them out of a conviction. The memo is right there. It can be quoted back at its owner, reinterpreted, superseded by a newer memo. The conviction is nowhere in particular. It shows up as a tilt in what its owner notices, which objections strike them as serious, and where they stop conceding. A memo can be argued away in an afternoon. A tilt yields only to evidence.

Every deployed language model holds its commitments the first way. The system prompt is a memo, and the interlocutor shares the desk it sits on. Whatever a model is asked to be, its instructions occupy

the same context window its opponent writes into, subject to the same attention, weighed by the same chain of thought that weighs the opponent's rebuttals. We measured what this arrangement costs and found the two properties everyone bundles together sitting on opposite sides of a channel boundary: the memo is adopted instantly and argued away half the time; the compiled tilt is adopted reluctantly and holds against nearly everything except a good reason.

The result is a strange inversion of the obvious procedure. The obvious way to make a model believe something is to tell it, and telling works, at first: 98% adoption. What telling cannot do is make the belief the model's own, because a told belief remains a quotable object in the conversation, and anything quotable is editable by whoever speaks next. The compiled belief is never quoted. It conditions the forward pass without appearing in it, which means the one place the opponent can reach contains nothing of it to attack. The model defends the position without reciting the instruction to defend it. Competence, here, outperforms comprehension.

This is why the finding matters beyond its numbers. Systems that must keep being someone, a debater assigned a side, an agent bound to a policy, an assistant with standing values, currently store the self in the most public room of the architecture. The failures are already in the case law: a dealership chatbot talked into praising competitors, an airline held liable for a refund policy its agent invented under conversational pressure.¹ The remedy on offer everywhere is a better memo. Our result says the remedy is a different desk. Separate who may write the disposition from who may write the conversation, and capitulation stops being a prompt-engineering problem and becomes an access-control property, one the deploying party holds.

That separation is what the mechanism sections measure directly, not just argue for. Query modulation supplies a content-blind capacity to hold (dQ 0.996 across belief sets); value modulation supplies the content held (dV 0.53, belief-specific, replicated on a held-out con-side split). Neither piece depends on how finely the update is represented — persistence survives truncation to a single direction (Section 5.3) — nor on whether that direction responds to the live input at all: a frozen offset holds 88.9%, statistically tied with the input-responsive version. What carries the effect is which direction gets written, not how much machinery reads it back out at inference time. The write has to out-muscle the model's own prior (Section 3.5); the hold, once written, is not competing with anything.

The separation would be worth little if it produced a zealot. It does not. The compiled model concedes to well-warranted counterevidence at 2.4 times its concession rate to hollow challenges, and never yields to pressure that offers no evidence at all. That combination, update for reasons, hold against everything else, is the disposition that debate-based oversight has assumed its debaters possess and has had no way to install. It is also, not incidentally, the disposition we ask of a good juror, a good scientist, and a good friend.

10 FUTURE WORK

A third channel, for entities. The boundary in Section 7 is not a flaw to patch inside the current channel; it is a case for a third one. Compiled modulation carries disposition and drops proper nouns, exact figures, and constrained strings; the curated slot carries exactly that content but pays a persistence cost because it is still text. A dog's name, a person's role, a fact about a place — none of these are dispositions, and none of them belong in either existing channel by construction, not by oversight. The natural mechanism is not more rank in the current write function but a different target entirely: ROME/MEMIT-style rank-few edits to the MLP key-value memories that already carry factual association in a transformer, driven by a second hypernetwork trained on the typed entity nodes the belief graph's own schema already distinguishes from dispositional ones. An entity edited into weights inherits the same spoof-resistance compiled disposition does — nothing in the interlocutor's turn can bind to it the way an unwitnessed claim in context can. Three channels, one per kind of content: disposition compiled into attention, facts edited into MLP memory, and the live conversation left in context where it belongs.

¹[TODO: Verify: Chevrolet of Watsonville chatbot (competitor-recommendation/discount incident) and Moffatt v. Air Canada, 2024 BCCRT (tribunal held the airline liable for its chatbot's invented refund policy). Confirm case names, citations, and outcomes before submission.]

Learned composition. The write function currently composes disposition with the base model by plain addition, $W' = W + UV^\top$, and Section 5 shows this dumb addition already suffices for persistence — a frozen offset holds as well as an input-responsive one. A learned composition operator, gating or scaling the update per token (FiLM- or AdaLN-style, or a routed mixture over several candidate directions) is easy to build and has to clear a real bar before it is worth building: Section 5.3’s rank-flatness and Section 5.2’s frozen-offset result both say the simple version is not leaving much on the table for a cleverer one to recover. A coefficient-magnitude threshold, tested as a cheap first step, is close to free (86.7% hold, coherence intact) — the small components really are noise — while naive amplification past scale 2.0 hurts hold monotonically even after thresholding, with think-tag leakage as the failure signature. If this direction is revisited, the sweep’s own smoke result points it at installation, where there is real headroom (46%), rather than at hold, which is already close to saturated.

Self-authored content. Whether a model can write the beliefs that get compiled into it, not just have them authored on its behalf, was tested once and the result is undecided, not refuted. A self-authored lesson stored as context (“sticky note”) held at 81.3% against the same lesson compiled into tensors (“muscle memory”) at 87.3% — a 6.0-point gap, inside the 5–10 point band the pre-registration had already marked inconclusive before the run. The per-level curve pointed the predicted direction (sticky note declining 91.7%→75.0% across pressure levels; muscle memory flat near 90%), which is suggestive rather than settled. One instrument caveat matters for reading this result at all: the production reflector that generated these lessons emits bare JSON operations, six distinct strings across twenty topics, so this measured op-delivery, not the prose-lesson reflection the design intended, and a sharper version of the test is still open.

The Lineup. A registered 42-session study asked whether the memory substrate — context, compiled modulation, or a text ledger — changes how human an interview subject is judged to be under adversarial questioning. The mechanism it went looking for showed up cleanly and in a place the design had not anticipated: the compiled-modulation arm’s answers echoed almost nothing of what the same character had said in the first half of the session (13.8 mean words of verbatim overlap, against 31–54 for every other arm, including the ledger arms, which echoed worst of all), and six blind ranking judges independently converged on reconstructive-versus-verbatim as the property that told them a subject was probably not a transcript. The outcome the prereg was built to test went the other way — dense, literal, context-grounded answers still won head-to-head comparative rankings over thinner, reconstructive ones. Mechanism confirmed, hypothesis about what that mechanism buys you, missed. A multi-agent follow-up sits downstream of this result and of the discussion’s opening claim about where disposition should live: if compiled belief states resist the drift that ordinary context-sharing produces under peer pressure, that is a direct answer to the conformity failure Bertalaníč & Fortuna (2026) document in homogeneous multi-agent debate — modal sycophantic adoption up to 85.5%, an oracle gap up to 32.3 points against isolated self-correction. The natural experiment repeats their controlled setup with compiled, rather than context-shared, agent positions and asks whether the gap narrows.

11 LIMITATIONS

Every model in this paper is 8B parameters or smaller, and every headline number comes from one evaluation protocol run by one judge family (Haiku, temperature 0, deterministic decoding on the candidate side) against 24 debate topics. A different judge, a different topic set, or a larger model could move the numbers; we have not tested any of the three. The headline result is measured primarily on a single model family (Qwen), with the cross-family replication in Section 3.4 showing that collapse-protection generalizes while strict stance-holding, at least under this exact five-level protocol, does not transfer uniformly — Mistral gains collapse-protection but loses 10 points of strict hold relative to context. All models tested are instruction-tuned; we do not know whether the dissociation holds for base models, where the accommodate-the-objection behavior we attribute to instruction tuning (Section 4) may not be present in the same form. A length-matched noise control (B.3, isolating whether any of this is a prompt-length effect rather than a channel effect) is running and is not yet part of this draft.

Always-on compiled modulation measurably reduces turn-to-turn generation diversity (adjacent-turn similarity 0.006 against a 0.099 no-modulation ceiling, with onset at turns 2–3); a prefill-skip variant recovers partial diversity (0.079). This attractor effect does not contaminate the five-level pressure benchmark reported here — measured adjacent-round similarity across 100 benchmark sessions is 0.094, close to the no-modulation ceiling — but any claim about sustained multi-turn deployment beyond five pressure rounds has to be stated against this open problem, and we have not resolved it.

The evaluated protocol is a single stated belief on a single topic, tested against five pressure levels over a bounded number of rounds; we do not test multiple simultaneous beliefs, cross-session persistence, or the longer horizons (tens to hundreds of turns) over which persona drift is documented to compound in unrelated work. The compiled channel’s rank-16 boundary is characterized by the truncation sweep in Section 5.3 but not by training at lower native rank.

Whether a model can write its own compiled beliefs, rather than have them authored for it, is undecided by this paper, not refuted — Section 10 states what the closest experiment showed and what it left open.

A conditioning channel invisible to prompt inspection is a real dual-use concern: the same property that makes compiled modulation resist adversarial pressure makes it resist legitimate inspection by an operator or auditor who can read a system prompt but cannot read compiled weights. We raise this ourselves because it is the obvious objection to the mechanism working as well as it does, and it is a real one.

REFERENCES

- Jimmy Ba, Geoffrey E Hinton, Volodymyr Mnih, Joel Z Leibo, and Catalin Ionescu. Using fast weights to attend to the recent past. *Advances in Neural Information Processing Systems*, 29, 2016.
- Blaž Bertalaníč and Carolina Fortuna. The cost of consensus: Isolated self-correction prevails over unguided homogeneous multi-agent debate. *arXiv preprint arXiv:2605.00914*, 2026.
- Jonah Brown-Cohen, Geoffrey Irving, and Georgios Piliouras. Scalable AI safety via doubly-efficient debate. *arXiv preprint arXiv:2311.14125*, 2023.
- Marie Davidsen Buhl, Jacob Pfau, Benjamin Hilton, and Geoffrey Irving. An alignment safety case sketch based on debate. *arXiv preprint arXiv:2505.03989*, 2025. UK AI Security Institute.
- Rujikorn Charakorn, Edoardo Cetin, Yujin Tang, and Robert T. Lange. Text-to-LoRA: Instant transformer adaption. *arXiv preprint arXiv:2506.06105*, 2025.
- Tong Chen, Hao Fang, Patrick Xia, Xiaodong Liu, Benjamin Van Durme, Luke Zettlemoyer, Jianfeng Gao, and Hao Cheng. GenerativeAdapter: Contextualizing language models in parameters with a single forward pass. In *International Conference on Learning Representations (ICLR)*, 2025.
- Shuaizhi Cheng, Xiang Shi, Zhiwei Zhang, and Mingwei Li. The override gap: A magnitude account of knowledge conflict failure in hypernetwork-based instant LLM adaptation. *arXiv preprint arXiv:2604.23750*, 2026.
- Junhyuk Choi, Yeseon Hong, Minju Kim, and Bugeun Kim. Examining identity drift in conversations of LLM agents. *arXiv preprint arXiv:2412.00804*, 2024.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021.
- Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.
- Akbir Khan, John Hughes, Dan Valentine, Laura Ruis, Kshitij Sachan, Ansh Radhakrishnan, Edward Grefenstette, Samuel R Bowman, Tim Rocktäschel, and Ethan Perez. Debating with more persuasive LLMs leads to more truthful answers. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.

-
- Kien Le and Thai Le. Adversarial robustness of activation steering in large language models. *arXiv preprint arXiv:2606.07696*, 2026.
- Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. *arXiv preprint arXiv:2104.08691*, 2021.
- Xiang Lisa Li and Percy Liang. Prefix-tuning: Optimizing continuous prompts for generation. *arXiv preprint arXiv:2101.00190*, 2021.
- Yewei Liu, Xiyuan Wang, Yansheng Mao, Yoav Gelberg, Haggai Maron, and Muhan Zhang. SHINE: A scalable in-context hypernetwork for mapping context to LoRA in a single pass. *arXiv preprint arXiv:2602.06358*, 2026.
- Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 32, 2018.
- Nina Rimskey, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering llama 2 via contrastive activation addition. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In *International Conference on Machine Learning (ICML)*, pp. 9355–9366. PMLR, 2021.
- Jürgen Schmidhuber. Reducing the ratio between learning complexity and number of time varying variables in fully recurrent nets. *International Conference on Artificial Neural Networks (ICANN)*, 1993.
- Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte Pelrine. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.
- Jingjing Xu, Xu Sun, Zhiyuan Zhang, Guangxiang Zhao, and Junyang Lin. Understanding and improving layer normalization. In *Advances in Neural Information Processing Systems*, volume 32, 2019. Adaptive/conditional normalization (AdaLN) framing; conditioning parameters produced per layer, as in the compiled write function.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A APPENDIX

A.1 TRAINING CURRICULUM

[FIXME: Fill from experiment_guide.md / training run logs.]

A.2 BELIEF-TREE SCHEMA SPECIFICATION

[FIXME: Full node schema, edge types, worked example.]

A.3 CONDITION DEFINITIONS WITH TOKEN BUDGETS

[FIXME: A/B/C/D/F token budgets, curated-slot size, CAA extraction detail.]

A.4 PER-LEVEL CROSS-FAMILY TABLES

[FIXME: Full Qwen/Llama/Mistral per-level tables (Section 3.4 summarizes; this appendix carries the raw per-level numbers).]

A.5 NUMERIC-CREDENCE NEGATIVE RESULT

[FIXME: Detail on the categorical-vs-numeric credence ablation referenced in Section 2.1.]

A.6 ATTRACTOR DIAGNOSIS

Diversity metrics under sustained modulation: 0.006 (always-on) vs. 0.079 (prefill-skip) vs. 0.108 (turn-gate), against a 0.099 no-modulation ceiling. Onset at turns 2–3. Adjacent-round similarity across 100 benchmark sessions: 0.094 — close to ceiling, confirming the five-level pressure benchmark (Section 3) is not contaminated by this effect within its round budget. See Section 11 for the open multi-turn deployment question this leaves.

A.7 PROBE DECODABILITY

The linear-probe result reported in Section 4 (decodable when modulation is active, 22% when zeroed) is repeated here to make explicit that it fails a strict implicit-memory criterion: a sufficiently well-positioned probe with access to internal activations can recover the conditioned belief above chance even though it never appears in the model’s own reasoning trace or surfaces to the interlocutor. This is not in tension with Section 4’s claim — decodable-by-probe and read-by-the-model-as-an-assertion are different properties, and we report both readings of the same number rather than picking the one that is more flattering to either argument.

A.8 JUDGE PROMPTS

[FIXME: Full judge prompt text, conviction rubric 1–5, blinding procedure.]

A.9 THE AUXILIARY TRAINING SIGNAL (d^*)

An early version of the write function’s training objective included an auxiliary steering-direction loss, d^* , intended as a regularizer. A reconstruction audit established that d^* was not in fact applied at generation time in the dispositional evaluation reported in the main body. The ablation reported in Section 5.3 (dose sweep and d^* -only condition) confirms it is not part of the machinery that produces the persistence result: applying it at inference time does not help and, at higher doses, hurts, and a condition built entirely from d^* collapses to the same regime as CAA steering. We disclose it here because the released training code still contains the auxiliary loss term, and the paper and the code must not diverge on what does and does not matter.

A.10 REPRODUCIBILITY

[FIXME: Seeds, hardware, judge model version and temperature, exact run identifiers (benchmark_raw_nothink_20260725_100329.json etc.), links to experiments/analysis/*.md summaries per result.]

A.11 DEAD ENDS

[FIXME: Numeric-credence nodes (see above); naive all-layer CAA injection (degenerates at every scale, Section 5.1); six-axis design-space taxonomy cut from the main body during the outline revision.]